

Assignment #3

Media Queries + Bootstrap Grid

Campus Life student guide

Team name	Campus Life
Team members	Yessetaiuly Inal; Nurkeldi Baizhigit
Group	SE-2508
Repository	github.com/inalesetaiuly/web-technology_project
Deployed website	To be added after GitHub Pages or Netlify deployment
Report date	29 September 2026

Submission note: the group number and deployed website URL were not available in the local project, so they are left as explicit fields to complete before submission.

1. Objective and project overview

The goal of this assignment is to create responsive web pages using CSS media queries and the Bootstrap grid system. The Campus Life website helps students find study tips, campus resources, student activities and contact information.

The project contains four linked pages: Home, About Us, Resources and Contact. The same responsive navigation is used throughout the site, while each page demonstrates a different layout or Bootstrap component.

2. Compliance summary

Responsive typography	Base, tablet and desktop font sizes in <code>css/style.css</code> .	Implemented
Media-query card group	Three custom <code>.media-card</code> elements with mobile, tablet and desktop breakpoints.	Implemented
Bootstrap grid	<code>container</code> , <code>row</code> , <code>col-lg-6</code> and <code>col-lg-4</code> classes on multiple pages.	Implemented
Bootstrap components	<code>Navbar</code> , <code>buttons</code> , <code>carousel</code> , <code>cards</code> , <code>form</code> , <code>input-group</code> and <code>btn-group</code> .	Implemented
Accessibility	Semantic elements, <code>alt</code> text, <code>labels</code> , <code>captions</code> , <code>scope</code> and <code>ARIA</code> attributes.	Implemented
Report and ZIP	This report is generated; project ZIP still needs to be prepared.	Pending
Deployment URL	No deployed URL is present in the project yet.	Pending

3. Part 1 - CSS Media Queries

Task 1 - Responsive typography

The stylesheet uses a mobile-first base size, then changes the page title, h2 headings and paragraph text at 768px and 992px. This creates separate mobile, tablet and desktop typography scales.

Task 2 - Three-card responsive group

The Explore Campus Life section contains three custom media cards. The group uses flex-wrap and CSS media queries only: one card per row below 576px, two cards per row from 576px to 991.98px, and three cards per row from 992px.

```
css/style.css - responsive card rules
131 /* Part 1, Task 2: this card group uses CSS media queries instead of Bootstrap columns. */
132 .media-card-container {
133   display: flex;
134   flex-wrap: wrap;
135   gap: 1.5rem;
136   align-items: stretch;
137 }
138
139 .media-card {
140   display: flex;
141   flex: 1 1 300px;
142   flex-direction: column;
143   min-width: 0;
144   padding: 1rem;
145   background-color: var(--surface);
146   border: 1px solid var(--border);
147   border-radius: 0.5rem;
148   box-shadow: 0 4px 16px rgba(11, 37, 69, 0.06);
149   transition: transform 0.3s ease, box-shadow 0.3s ease;
150 }
151
152 .media-card img {
153   width: 100%;
154   height: 11.25rem;
155   object-fit: cover;
156   border-radius: 0.5rem;
157 }
158
159 .media-card h3 {
160   color: var(--nav-000);
161   margin-top: 1rem;
162   margin-bottom: 0.625rem;
163 }
164
165 .media-card p {
166   margin-bottom: 1.25rem;
167 }
168
169 .media-card-button {
170   display: inline-block;
171   margin-top: auto;
172   padding: 0.625rem 0.9375rem;
173   background-color: var(--nav-000);
174   border-radius: 0.3125rem;
175   color: #ffffff;
176   text-align: center;
177   text-decoration: none;
178 }
179
180 .media-card-button:hover,
181 .media-card-button:focus {
182   background-color: var(--nav-700);
183   color: #ffffff;
184 }
185
186 .media-card-hover {
187   box-shadow: 0 16px 22px rgba(11, 37, 69, 0.16);
188   transform: translateY(-0.3125rem);
189 }
```

Figure 1. Code evidence for responsive typography and the custom media-card breakpoints.

4. Part 2 - Bootstrap implementation

Task 3 - Bootstrap grid

The project uses .container and .row elements with responsive columns. The Home and Resources pages use two col-lg-6 columns for larger screens. The Resources page uses three col-lg-4 columns for the featured resource cards, with col-12 and col-md-6 fallbacks for smaller screens.

Task 4 - Bootstrap spacing utilities

Layout spacing is handled with Bootstrap utilities such as py-4, p-4, mb-4, mb-5, g-4, mt-4, ms-auto and responsive spacing classes. Custom spacing remains only where it is part of the custom media-card styling or visual theme.

```

resources.html – Bootstrap grid and cards
70 <section class="mb-4" aria-labelledby="featured-resources-title">
71 <h2 id="featured-resources-title" class="mb-4">Featured Resources</h2>
72 <div class="row g-4">
73 <div class="col-12 col-md-6 col-lg-4">
74 <article class="card resource-card h-100">
75 
77 <h3 class="card-title h5">Library</h3>
78 <p class="card-text">Find books, quiet study spaces and help with academic r
79 <a href="contact.html" class="btn btn-primary mt-auto">Ask the Library</a>
80 </div>
81 </article>
82 </div>
83
84 <div class="col-12 col-md-6 col-lg-4">
85 <article class="card resource-card h-100">
86 
88 <h3 class="card-title h5">Study Spaces</h3>
89 <p class="card-text">Choose a comfortable place to focus on individual or gr
90 <a href="contact.html" class="btn btn-outline-primary mt-auto">Find Support<
91 </div>
92 </article>
93 </div>
94
95 <div class="col-12 col-md-6 col-lg-4">
96 <article class="card resource-card h-100">
97 
99 <h3 class="card-title h5">Student Center</h3>
100 <p class="card-text">Discover activities, clubs and opportunities to meet ot
101 <a href="about.html" class="btn btn-secondary mt-auto">Explore Activities</a>
102 </div>
103 </article>

```

Figure 2. Code evidence for the Bootstrap grid and the three resource cards.

Task 5 - Bootstrap navbar

All four pages use a responsive Bootstrap navbar with four links. The navbar-toggler button controls the collapsed menu below the large breakpoint and includes aria-controls, aria-expanded and aria-label attributes.

Task 6 - Bootstrap buttons

The project demonstrates btn-primary, btn-secondary, btn-outline-primary, btn-outline-secondary, btn-lg and btn-group variants on the Home, Resources and Contact pages.

Task 7 - Bootstrap carousel

About Us contains the campusCarousel component with nine carousel items, nine indicators, previous and next controls, captions and descriptive alternative text.

```

about.html – Bootstrap carousel
100 <section class="section-panel p-0" aria-labelledby="moments-title">
101 <h2 id="moments-title" class="mb-4">Campus Moments</h2>
102
103 <div id="campusCarousel" class="carousel slide campus-carousel" aria-label="Campus photo gallery">
104 <div class="carousel-indicators">
105 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="0" class="active"
106 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="1"
107 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="2"
108 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="3"
109 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="4"
110 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="5"
111 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="6"
112 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="7"
113 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="8"
114 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="9"
115 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="10"
116 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="11"
117 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="12"
118 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="13"
119 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="14"
120 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="15"
121 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="16"
122 <button type="button" data-bs-target="#campusCarousel" data-bs-slide-to="17"
123 </div>
124
125 <div class="carousel-inner rounded">
126 <div class="carousel-item active">
127 
128 <div class="carousel-caption d-block"><h3 class="h5 mb-0">Group Study</h3></div>
129 </div>
130 <div class="carousel-item">
131 
132 <div class="carousel-caption d-block"><h3 class="h5 mb-0">University Library</h3></div>
133 </div>
134 <div class="carousel-item">
135 
136 <div class="carousel-caption d-block"><h3 class="h5 mb-0">Student Activities</h3></div>
137 </div>
138 <div class="carousel-item">
139 
140 <div class="carousel-caption d-block"><h3 class="h5 mb-0">Computer Lab</h3></div>
141 </div>
142 <div class="carousel-item">
143 
144 <div class="carousel-caption d-block"><h3 class="h5 mb-0">Campus Life</h3></div>
145 </div>
146 <div class="carousel-item">
147 
148 <div class="carousel-caption d-block"><h3 class="h5 mb-0">Team Projects</h3></div>
149 </div>
150 <div class="carousel-item">
151 
152 <div class="carousel-caption d-block"><h3 class="h5 mb-0">Campus Events</h3></div>
153 </div>
154 <div class="carousel-item">
155 

```

Figure 3. Code evidence for the nine-slide Bootstrap carousel.

Task 8 - Bootstrap cards

Resources contains three Bootstrap cards. Each card has a card-img-top image, card-body, card-title, card-text and button, and the cards are placed in a responsive Bootstrap row.

Task 9 - Responsive form

Contact contains a responsive form with form-control inputs, a form-select, an input-group for email, a textarea, a checkbox and a btn-group with submit and reset buttons.

```
180 <form action="#" method="post">
181   <div class="row g-3">
182     <div class="col-12 col-md-6">
183       <label class="form-label" for="fullname">Full Name</label>
184       <input class="form-control" type="text" id="fullname" name="fullname"
185         placeholder="Enter your full name" autocomplete="name" required>
186     </div>
187     <div class="col-12 col-md-6">
188       <label class="form-label" for="email">Email</label>
189       <div class="input-group">
190         <span class="input-group-text" aria-hidden="true">@</span>
191         <input class="form-control" type="email" id="email" name="email"
192           placeholder="Enter your email" autocomplete="email" required>
193       </div>
194     </div>
195     <div class="col-12 col-md-6">
196       <label class="form-label" for="subject">Subject</label>
197       <input class="form-control" type="text" id="subject" name="subject"
198         placeholder="Enter subject" required>
199     </div>
200     <div class="col-12 col-md-6">
201       <label class="form-label" for="category">Topic / Department</label>
202       <select class="form-select" id="category" name="category" required>
203         <option value="" selected disabled>Select an option ---</option>
204         <option value="general">General Question</option>
205         <option value="it_support">IT Support</option>
206         <option value="library">Library Resources</option>
207         <option value="academic">Academic Advisory</option>
208         <option value="feedback">Feedback</option>
209       </select>
210     </div>
211     <div class="col-12">
212       <label class="form-label" for="message">Message</label>
213       <textarea class="form-control" id="message" name="message" rows="5"
214         placeholder="Write your message here..." required></textarea>
215     </div>
216     <div class="col-12">
217       <div class="form-check">
218         <input class="form-check-input" type="checkbox" value="yes" id="followup" name="followup">
219         <label class="form-check-label" for="followup">I would like a reply by email.</label>
220       </div>
221     </div>
222     <div class="col-12">
223       <div class="btn-group" role="group" aria-label="Form actions">
224         <button class="btn btn-primary btn-lg" type="submit">Send Message</button>
225         <button class="btn btn-outline-secondary btn-lg" type="reset">Clear Form</button>
226       </div>
227     </div>
228   </div>
229 </form>
```

Figure 4. Code evidence for the responsive Bootstrap form.

Task 10 - Accessibility and semantic HTML

The pages use header, nav, main, section, article, aside and footer elements. Images have descriptive alt text, form fields have labels, tables have captions and scoped headers, and interactive components include appropriate ARIA attributes.

5. Visual page previews

The following source-based previews summarize the resulting page layouts and components. They are included as report evidence for the local project. For a final submission, replace them or supplement them with browser screenshots after the site is opened at the required screen sizes.

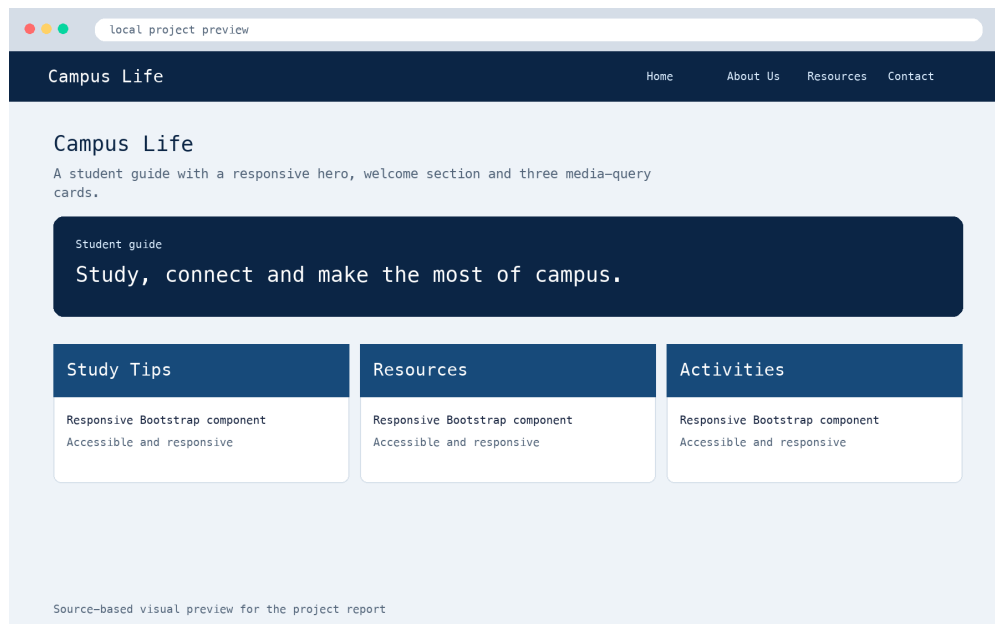


Figure 5. Home page preview.

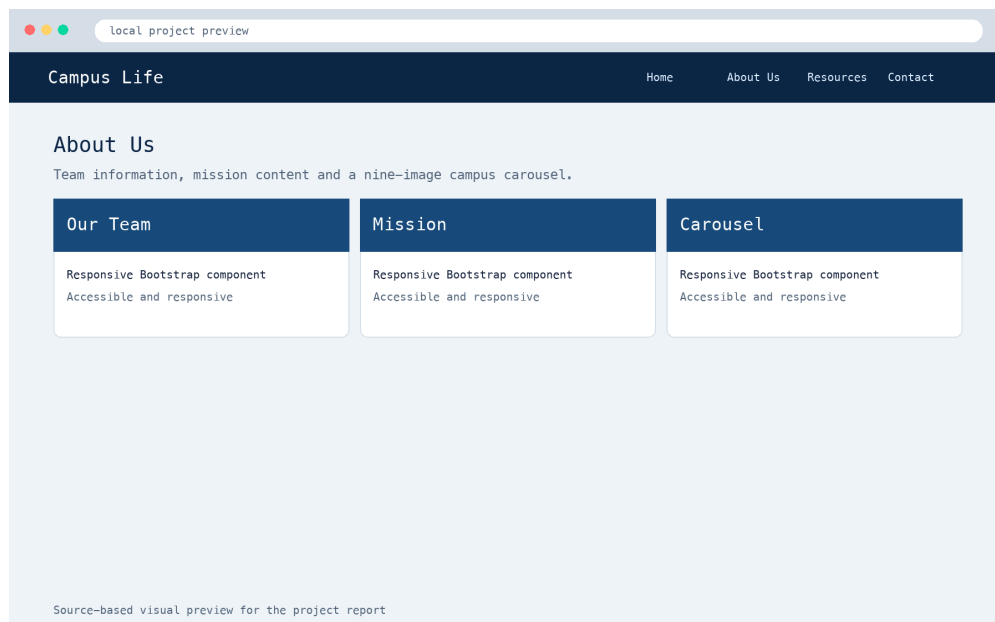


Figure 6. About Us page preview.

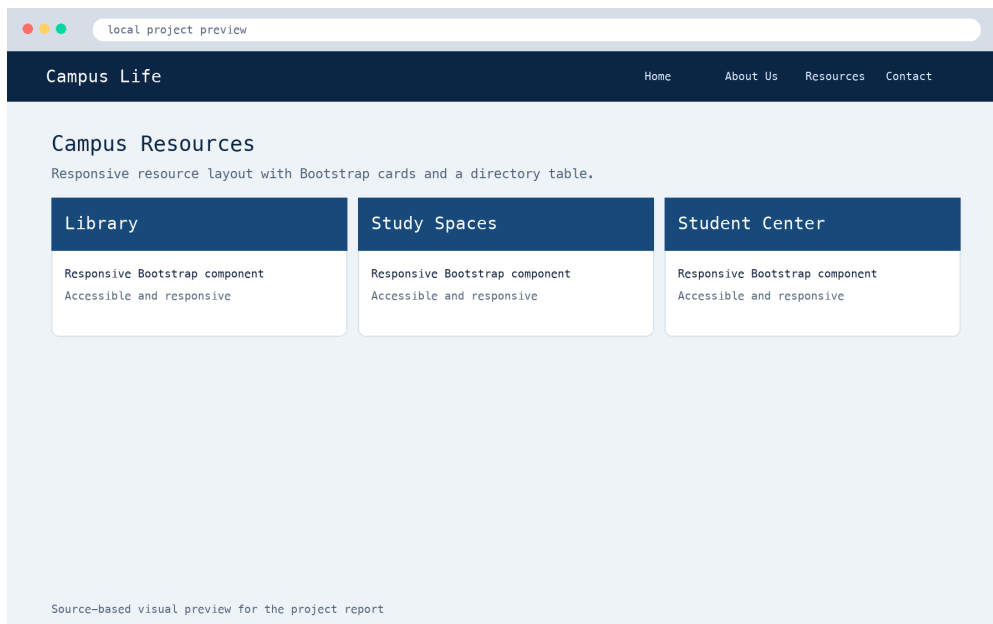


Figure 7. Resources page preview.

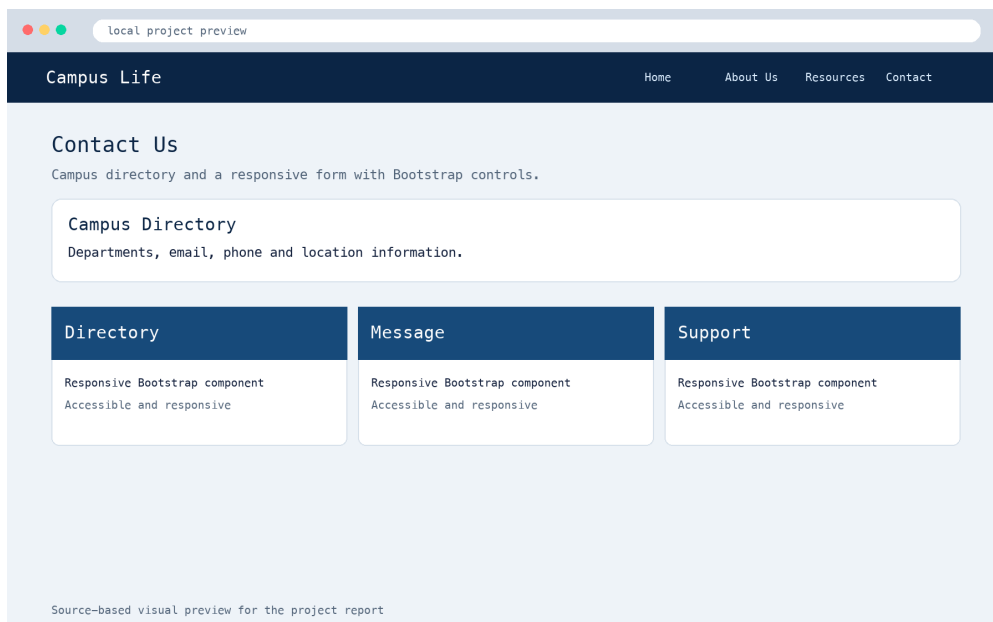


Figure 8. Contact page preview.

6. Step-by-step implementation

1. Kept the Campus Life content and four-page navigation structure.
2. Added Bootstrap 5.3.8 CSS and JavaScript CDN links to every HTML page.
3. Replaced the custom navigation with a responsive Bootstrap navbar and hamburger button.
4. Added mobile-first typography rules and media-query breakpoints for the three custom cards.
5. Converted the main layouts to Bootstrap containers, rows and responsive columns.
6. Added Bootstrap buttons, button groups, cards, carousel and form controls.
7. Added accessibility details including labels, alt text, captions, scope and ARIA attributes.
8. Checked HTML tag structure, duplicate IDs, local asset paths and git diff formatting.

7. Submission checklist

PDF report	Created as this file.
Project folder in ZIP format	Still needs to be created from the project folder.
Group number	Fill in the blank field on the cover page.
Deployed website URL	Add the GitHub Pages or Netlify URL after deployment.
Commit and push	The current five local changes still need to be committed and pushed.
Individual submission and defense	Complete separately through the LMS and practice lesson.

Conclusion

The local Campus Life project implements the required CSS media queries and Bootstrap components. The remaining submission actions are to fill in the group number, capture final browser screenshots, commit and push the changes, deploy the website, add the deployment URL, create the ZIP archive and submit the PDF and ZIP for every team member.